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# =====
# 20. Harris Corner Detection Parameter Study
# Google Colab Implementation
# =====

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# -----
# STEP 1: Upload Image
# -----
print("Upload an image")
uploaded = files.upload()

img_path = next(iter(uploaded))

# Read image
img = cv2.imread(img_path)
rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Convert to float32 for Harris
gray_f = np.float32(gray)

# -----
# Different parameter sets
# (blockSize, ksize, k)
# -----
params = [
    (2, 3, 0.04),
    (3, 3, 0.04),
    (5, 3, 0.04),
    (5, 5, 0.06)
]

# -----
# Display Results
# -----
```

```

fig, axes = plt.subplots(1, 4, figsize=(18,5))

for i, (blockSize, ksize, k) in enumerate(params):

    temp = rgb.copy()

    # Harris corner detection
    dst = cv2.cornerHarris(gray_f, blockSize, ksize, k)

    # Mark detected corners in red
    temp[dst > 0.01 * dst.max()] = [255, 0, 0]

    axes[i].imshow(temp)
    axes[i].set_title(f"b={blockSize}, ks={ksize}\\nk={k}")
    axes[i].axis('off')

plt.suptitle("Harris Corner Detection Parameter Study",
             fontsize=18, fontweight='bold')
plt.tight_layout()
plt.show()

```

Upload an image

img3.jpeg

img3.jpeg(image/jpeg) - 10583 bytes, last modified: 20/07/2026 - 100% done

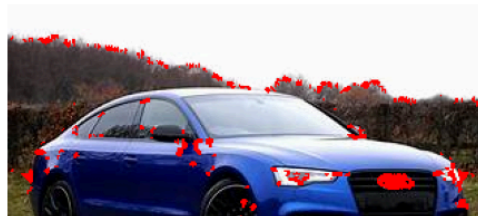
Saving img3.jpeg to img3.jpeg

Harris Corner Detection Parameter Study

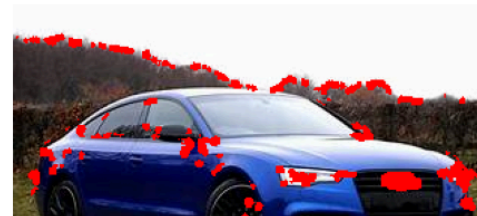
b=2, ks=3\\nk=0.04



b=3, ks=3\\nk=0.04



b=5, ks=3\\nk=0.04



b=5, ks=5\\nk=0.06



